

HAOZHE (TONY) CHEN

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EDUCATION

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|---|----------------------------|
| Princeton University <ul style="list-style-type: none">- Ph.D., Computer Science- Advisor: Zhuang Liu | <i>Aug 2025 - May 2030</i> |
| Columbia University, Columbia Engineering <ul style="list-style-type: none">- B.S., Computer Science, GPA: 4.05/4 | <i>Aug 2021 - Dec 2024</i> |

RESEARCH EXPERIENCE

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| Decision, Risk, and Operations Division, Columbia Business School <i>Researcher</i> <ul style="list-style-type: none">- Researched principled frameworks for understanding LLM pretraining data mixture- Researched simulation frameworks for queueing networks and foundation model serving- Researched simulating human behaviors with Large Language Model- Advisor: Hongseok Namkoong, Tianyi Peng | <i>Apr 2023 - Present</i> |
| Computer Vision Lab, Columbia University <i>Researcher</i> <ul style="list-style-type: none">- Researched robustness and interpretability of vision and language models- Researched adaptive evaluation of foundation models- Researched model architectures with flexible inference-time compute scaling- Advisor: Carl Vondrick, Chengzhi Mao | <i>Sep 2022 - Present</i> |
| Spoken Language Processing Group, Columbia University <i>Researcher</i> <ul style="list-style-type: none">- Researched controllable synthesis of emotional speech- Built datasets for empathetic speech and conducted acoustic-lexical analysis- Advisor: Julia Hirschberg | <i>Nov 2021 - Present</i> |
| Sen Lab, UC Santa Barbara <i>Research Intern</i> <ul style="list-style-type: none">- Researched 3D object reconstruction model with VAE and 3D-GAN | <i>June 2020 - Aug 2020</i> |

INDUSTRY EXPERIENCE

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| Penrose Lab <i>Part Time Researcher</i> <ul style="list-style-type: none">- Building voice agent for healthcare | <i>Sep 2025 - Present</i> |
| Together AI <i>Research Intern</i> <ul style="list-style-type: none">- Researched LLM model routing, approximate speculative decoding, and new speculative decoding architecture | <i>Sep 2024 - Aug 2025</i> |
| Voloridge Investment Management <i>Quantitative Research Intern</i> <ul style="list-style-type: none">- Researched equity factor modeling with neural network- Researched predicting stock return with large-scale natural language datasets | <i>May 2023 - Dec 2023</i> |
| Voloridge Investment Management <i>Quantitative Research Intern</i> <ul style="list-style-type: none">- Researched stock market clustering based on price history and corporation relations using graph neural network- Produced stock groupings with improved intra-group correlations and adaptability to market changes | <i>June 2021 - Aug 2021</i> |

PUBLICATIONS

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- Ang Li, **Haozhe Chen**, Hongseok Namkoong, Tianyi Peng. LLM Generated Persona is a Promise with a Catch. *Advances in Neural Information Processing Systems* 38, 2025.
 - Olivier Toubia, G. Z. Gui, Tianyi Peng, Daniel J. Merlau, Ang Li, **Haozhe Chen**. Twin-2K-500: A Data Set for Building Digital Twins of over 2,000 People Based on Their Answers to over 500 Questions. *Marketing Science*, 2025.
 - Ta-Chung Yen, Andrew W. T. Siah, **Haozhe Chen**, Tianyi Peng, Daniel Guetta, Hongseok Namkoong. Data Mixture Optimization: A Multi-fidelity Multi-scale Bayesian Framework. *Advances in Neural Information Processing Systems* 38, 2025.
 - **Haozhe Chen**^{*}, Ang Li^{*}, Ethan Che, Jing Dong, Tianyi Peng, Hongseok Namkoong. “QGym: Scalable Simulation and Benchmarking of Queuing Network Controllers.” *Advances in Neural Information Processing Systems* 37, 2024.
 - **Haozhe Chen**, Run Chen, Julia Hirschberg. “EmoKnob: Enhance Voice Cloning with Fine-Grained Emotion Control.” *arXiv preprint arXiv:2410.00316*, 2024.
 - **Haozhe Chen**, Carl Vondrick, Chengzhi Mao. “SelfIE: Self-Interpretation of Large Language Model Embeddings.”

arXiv preprint arXiv:2403.10949, 2024.

- **Haozhe Chen**, Junfeng Yang, Carl Vondrick, Chengzhi Mao. “INVITE: INterpret and Control Vision-Language Models with Text Explanations.” *The Twelfth International Conference on Learning Representations*, 2024.
- Run Chen, **Haozhe Chen**, Anushka Kulkarni, Eleanor Lin, Linda Pang, Divya Tadimeti, Julia Hirschberg. “Detecting Empathy in Speech.” *Proc. INTERSPEECH*, 2024.

TEACHING EXPERIENCE

Columbia Business School *Teaching Assistant*

July 2024 - Present

- B9153 Generative AI: Technical and Social (PhD Course). Instructor: [Tianyi Peng](#), [Hannah Li](#).
- Assisted course material development on building and training Large Language Model.

Columbia University *Teaching Assistant*

Jan 2023 - May 2023

- COMS 3261 Computer Science Theory. Instructor: [Xi Chen](#).
- Held office hours, graded homeworks & exams.

SERVICE

Reviewer

- NeurIPS 2024 / 2025
- ICLR 2025
- AISTATS 2025
- ICML 2025
- ACL 2025

HONORS AND AWARDS

Kaggle Competition Expert (*#144 of 160,000+*)

2020

- Designed & trained models in natural language processing & computer vision in 20+ Kaggle competitions. Peak rank #144 out of 160,000+ users (top 0.1%)
- Won 7 silver medals and 9 bronze medals. Details available on Kaggle profile: kaggle.com/tonychenxyz